

Web Development

60 Practicals — All Codes

VSE-281-COM

Second Year Computer Engineering (2024 Pattern)

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Practical No. 1

Q. Create an HTML page to display a student resume using headings, paragraphs, and lists.

```
<!DOCTYPE html>
<html>
<head><title>Student Resume</title></head>
<body>
  <h1>Shreyas Patil</h1>
  <p>Email: shreyas@mail.com | Phone: 9876543210</p>
  <hr>
  <h2>Education</h2>
  <ul>
    <li>B.E. Computer Engineering - SPPU (2023-2027)</li>
    <li>HSC - Pune Board - 82%</li>
    <li>SSC - Pune Board - 90%</li>
  </ul>
  <h2>Skills</h2>
  <ul>
    <li>HTML, CSS, JavaScript</li>
    <li>Python, Java</li>
    <li>MySQL</li>
  </ul>
  <h2>Hobbies</h2>
  <p>Cricket, Coding, Reading</p>
</body>
</html>
```

Practical No. 2

Q. Design a web page containing an image gallery, hyperlinks to social media pages, and a timetable using tables.

```
<!DOCTYPE html>
<html>
<head><title>Gallery and Timetable</title></head>
<body>
  <h2>Image Gallery</h2>
  
  
  

  <h2>Social Media</h2>
  <a href="https://www.instagram.com" target="_blank">Instagram</a> |
  <a href="https://www.twitter.com" target="_blank">Twitter</a> |
  <a href="https://www.youtube.com" target="_blank">YouTube</a>

  <h2>Timetable</h2>
  <table border="1" cellpadding="8">
    <tr><th>Day</th><th>9-10</th><th>10-11</th><th>11-12</th></tr>
    <tr><td>Monday</td><td>DBMS</td><td>DM</td><td>IOT</td></tr>
    <tr><td>Tuesday</td><td>COM</td><td>DBMS</td><td>DM</td></tr>
    <tr><td>Wednesday</td><td>IOT</td><td>COM</td><td>DBMS</td></tr>
  </table>
</body>
</html>
```

Practical No. 3

Q. Create an online feedback form with fields for username, phone number, course selection, comments, and submit button.

```
<!DOCTYPE html>
<html>
<head><title>Feedback Form</title></head>
<body>
  <h2>Feedback Form</h2>
  <form>
    <label>Username:</label><br>
    <input type="text" name="username" required><br><br>

    <label>Phone Number:</label><br>
    <input type="tel" name="phone" required><br><br>

    <label>Course:</label><br>
    <select name="course">
      <option>DBMS</option>
      <option>Web Development</option>
      <option>IOT</option>
      <option>Discrete Mathematics</option>
    </select><br><br>

    <label>Comments:</label><br>
    <textarea name="comments" rows="4" cols="40"></textarea><br><br>

    <input type="submit" value="Submit">
  </form>
</body>
</html>
```

Practical No. 4

Q. Apply CSS styles using internal, external, and inline methods to design a navigation menu and styled buttons.

```
<!DOCTYPE html>
<html>
<head>
  <title>CSS Navigation</title>
  <!-- Internal CSS -->
  <style>
    body { font-family: Arial; }
    nav { background-color: #333; padding: 10px; }
    nav a {
      color: white; text-decoration: none;
      margin: 0 15px; font-size: 16px;
    }
    nav a:hover { color: yellow; }
    .btn {
      padding: 10px 20px; border: none;
      border-radius: 5px; cursor: pointer;
      font-size: 14px; margin: 5px;
    }
  </style>
</head>
<body>
  <!-- Navigation Menu -->
  <nav>
    <a href="#">Home</a>
    <a href="#">About</a>
    <a href="#">Contact</a>
  </nav>

  <h2>Styled Buttons</h2>

  <!-- Inline CSS button -->
  <button class="btn" style="background-color: blue; color: white;">
    Primary
  </button>
  <button class="btn" style="background-color: green; color: white;">
    Success
  </button>
  <button class="btn" style="background-color: red; color: white;">
    Danger
  </button>
</body>
</html>
```

Practical No. 5

Q. Develop a responsive college website homepage using Bootstrap Navbar, Cards, and Grid System.

```
<!DOCTYPE html>
<html>
<head>
  <title>College Website</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
    rel="stylesheet">
</head>
<body>
  <nav class="navbar navbar-dark bg-dark">
    <div class="container">
      <span class="navbar-brand fw-bold">SPPU College</span>
    </div>
  </nav>

  <div class="container mt-4">
    <h2 class="text-center mb-4">Welcome to Our College</h2>
    <div class="row">
      <div class="col-md-4 mb-3">
        <div class="card shadow">
          <div class="card-body">
            <h5 class="card-title">Computer Engineering</h5>
            <p class="card-text">4 year B.E. program affiliated to SPPU.</p>
          </div>
        </div>
      </div>
      <div class="col-md-4 mb-3">
        <div class="card shadow">
          <div class="card-body">
            <h5 class="card-title">Mechanical Engineering</h5>
            <p class="card-text">4 year B.E. program affiliated to SPPU.</p>
          </div>
        </div>
      </div>
      <div class="col-md-4 mb-3">
        <div class="card shadow">
          <div class="card-body">
            <h5 class="card-title">Civil Engineering</h5>
            <p class="card-text">4 year B.E. program affiliated to SPPU.</p>
          </div>
        </div>
      </div>
    </div>
  </div>
</body>
</html>
```

Practical No. 6

Q. Write a JavaScript program to validate a registration form including password length, mobile number, and required fields.

```
<!DOCTYPE html>
<html>
<head>
  <title>Registration Validation</title>
  <script>
    function validate(e) {
      var name = document.getElementById("name").value;
      var mobile= document.getElementById("mobile").value;
      var pass = document.getElementById("pass").value;
      var msg = "";

      if (name == "")      msg += "Name is required.\n";
      if (!/^[0-9]{10}$/.test(mobile)) msg += "Enter valid 10-digit mobile.\n";
      if (pass.length < 6)  msg += "Password must be at least 6 characters.\n";

      if (msg != "") { alert(msg); e.preventDefault(); }
      else alert("Registration Successful!");
    }
  </script>
</head>
<body>
  <h2>Registration Form</h2>
  <form onsubmit="validate(event)">
    Name:  <input type="text" id="name"><br><br>
    Mobile: <input type="text" id="mobile"><br><br>
    Password:<input type="password" id="pass"><br><br>
    <input type="submit" value="Register">
  </form>
</body>
</html>
```

Practical No. 7

Q. Create a web page that changes background color and text dynamically using JavaScript DOM events.

```
<!DOCTYPE html>
<html>
<head><title>DOM Events</title></head>
<body>
  <h2 id="msg">Hello! Click a button.</h2>
  <br>
  <button onclick="changeRed()">Red</button>
  <button onclick="changeBlue()">Blue</button>
  <button onclick="changeGreen()">Green</button>
  <button onclick="reset()">Reset</button>

  <script>
    function changeRed() {
      document.body.style.backgroundColor = "red";
      document.getElementById("msg").innerHTML = "Background is Red!";
    }
    function changeBlue() {
      document.body.style.backgroundColor = "lightblue";
      document.getElementById("msg").innerHTML = "Background is Blue!";
    }
    function changeGreen() {
      document.body.style.backgroundColor = "lightgreen";
      document.getElementById("msg").innerHTML = "Background is Green!";
    }
    function reset() {
      document.body.style.backgroundColor = "white";
      document.getElementById("msg").innerHTML = "Hello! Click a button.";
    }
  </script>
</body>
</html>
```

Practical No. 8

Q. Write a JavaScript program to perform arithmetic operations using buttons and functions for a mini calculator.

```
<!DOCTYPE html>
<html>
<head>
  <title>Mini Calculator</title>
  <style>
    .box { border: 2px solid #333; padding: 20px; width: 260px; }
    input { width: 90%; margin-bottom: 8px; padding: 5px; }
    button{ width: 45%; padding: 8px; margin: 3px; cursor: pointer; }
    #res { font-weight: bold; color: blue; margin-top: 8px; }
  </style>
</head>
<body>
  <div class="box">
    <h3>Calculator</h3>
    <input type="number" id="n1" placeholder="Number 1">
    <input type="number" id="n2" placeholder="Number 2">
    <button onclick="calc('+')">Add</button>
    <button onclick="calc('-')">Sub</button>
    <button onclick="calc('*')">Mul</button>
    <button onclick="calc('/')">Div</button>
    <div id="res">Result: </div>
  </div>
  <script>
    function calc(op) {
      var a = parseFloat(document.getElementById("n1").value);
      var b = parseFloat(document.getElementById("n2").value);
      var r;
      switch(op) {
        case '+': r = a + b; break;
        case '-': r = a - b; break;
        case '*': r = a * b; break;
        case '/': r = b != 0 ? a / b : "Error: Div by 0"; break;
      }
      document.getElementById("res").innerHTML = "Result: " + r;
    }
  </script>
</body>
</html>
```

Practical No. 9

Q. Design a PHP script to display employee details and current server information.

```
<!DOCTYPE html>
<html>
<head><title>Employee Details</title></head>
<body>
  <h2>Employee Details</h2>
  <?php
    $name  = "Yash Sharma";
    $id    = 101;
    $dept  = "Computer Engineering";
    $salary = 50000;

    echo "Employee Name: " . $name . "<br>";
    echo "Employee ID: " . $id . "<br>";
    echo "Department: " . $dept . "<br>";
    echo "Salary: Rs. " . $salary . "<br><br>";

    echo "<h3>Server Information</h3>";
    echo "Server Name: " . $_SERVER['SERVER_NAME'] . "<br>";
    echo "PHP Version: " . phpversion() . "<br>";
    echo "Date & Time: " . date("d-m-Y H:i:s") . "<br>";
  ?>
</body>
</html>
```

Practical No. 10

Q. Write a PHP program to accept login credentials using POST method and display a successful login message.

```
<!DOCTYPE html>
<html>
<head><title>Login Form</title></head>
<body>
  <h2>Login Form</h2>
  <form method="POST">
    Username: <input type="text"      name="user" required><br><br>
    Password: <input type="password" name="pass" required><br><br>
    <input type="submit" name="btn" value="Login">
  </form>
  <hr>
  <?php
    if (isset($_POST['btn'])) {
      $user = htmlspecialchars($_POST['user']);
      echo "<h3 style='color:green;'>Login Successful!</h3>";
      echo "Welcome, <b>" . $user . "</b>!";
    }
  ?>
</body>
</html>
```

Practical No. 11

Q. Implement a PHP program for string operations such as uppercase conversion, lowercase conversion, and word count.

```
<!DOCTYPE html>
<html>
<head><title>String Operations</title></head>
<body>
  <h2>String Operations</h2>
  <?php
    $str = "Hello World from PHP";

    echo "Original: " . $str . "<br>";
    echo "Uppercase: " . strtoupper($str) . "<br>";
    echo "Lowercase: " . strtolower($str) . "<br>";
    echo "Word Count: " . str_word_count($str) . "<br>";
    echo "Length: " . strlen($str) . "<br>";
    echo "Reversed: " . strrev($str) . "<br>";
  ?>
</body>
</html>
```

Practical No. 12

Q. Create a PHP script to store student marks in an associative array and display them in tabular format.

```
<!DOCTYPE html>
<html>
<head><title>Student Marks</title></head>
<body>
  <h2>Student Marks</h2>
  <?php
    $marks = array(
      "Shreyas" => 85,
      "Yash"    => 90,
      "Tejas"   => 70,
      "Radha"   => 92,
      "Krishna" => 88
    );

    echo "<table border='1' cellpadding='8'>";
    echo "<tr><th>Student Name</th><th>Marks</th></tr>";
    foreach ($marks as $name => $m) {
      echo "<tr><td>$name</td><td>$m</td></tr>";
    }
    echo "</table>";
  ?>
</body>
</html>
```

Practical No. 13

Q. Create an HTML page to display details of a tourist place using headings, images, and formatted text.

```
<!DOCTYPE html>
<html>
<head><title>Tourist Place</title></head>
<body>
  <h1>Lonavala - A Beautiful Hill Station</h1>
  
  <h2>About</h2>
  <p>Lonavala is a hill station located in Pune district, Maharashtra.
    It is famous for its scenic beauty, waterfalls, and chikki.</p>
  <h2>Best Time to Visit</h2>
  <p><b>June to September</b> - Monsoon season is the most popular time.</p>
  <h2>Famous Attractions</h2>
  <ul>
    <li>Bhushi Dam</li>
    <li>Tiger's Leap</li>
    <li>Rajmachi Fort</li>
    <li>Karla Caves</li>
  </ul>
</body>
</html>
```

Practical No. 14

Q. Design a web page containing a nested list, hyperlinks to educational websites, and a table showing class schedules.

```
<!DOCTYPE html>
<html>
<head><title>Class Schedule</title></head>
<body>
  <h2>Subjects</h2>
  <ul>
    <li>Semester IV
      <ul>
        <li>DBMS</li>
        <li>Discrete Mathematics</li>
        <li>IOT</li>
        <li>COM</li>
      </ul>
    </li>
  </ul>

  <h2>Educational Links</h2>
  <a href="https://nptel.ac.in" target="_blank">NPTEL</a> |
  <a href="https://swayam.gov.in" target="_blank">Swayam</a> |
  <a href="https://unipune.ac.in" target="_blank">SPPU</a>

  <h2>Class Schedule</h2>
  <table border="1" cellpadding="8">
    <tr><th>Time</th><th>Monday</th><th>Tuesday</th><th>Wednesday</th></tr>
    <tr><td>9-10</td><td>DBMS</td><td>IOT</td><td>COM</td></tr>
    <tr><td>10-11</td><td>DM</td><td>DBMS</td><td>IOT</td></tr>
  </table>
</body>
</html>
```

Practical No. 15

Q. Create a job application form with fields for applicant name, qualification, experience, skills, and submit button.

```
<!DOCTYPE html>
<html>
<head><title>Job Application</title></head>
<body>
  <h2>Job Application Form</h2>
  <form>
    <label>Applicant Name:</label><br>
    <input type="text" name="name" required><br><br>

    <label>Qualification:</label><br>
    <select name="qualification">
      <option>B.E.</option>
      <option>M.E.</option>
      <option>Diploma</option>
    </select><br><br>

    <label>Experience (years):</label><br>
    <input type="number" name="exp" min="0"><br><br>

    <label>Skills:</label><br>
    <input type="checkbox" name="skill" value="html"> HTML
    <input type="checkbox" name="skill" value="css"> CSS
    <input type="checkbox" name="skill" value="js"> JavaScript
    <input type="checkbox" name="skill" value="php"> PHP<br><br>

    <input type="submit" value="Apply Now">
  </form>
</body>
</html>
```

Practical No. 16

Q. Apply CSS to design a webpage with styled forms, hover effects on buttons, and customized fonts.

```
<!DOCTYPE html>
<html>
<head>
  <title>CSS Styled Form</title>
  <style>
    body { font-family: Georgia, serif; background: #f0f0f0; }
    .form-box {
      background: white; padding: 25px;
      width: 350px; border-radius: 8px;
      box-shadow: 0 0 10px rgba(0,0,0,0.1);
    }
    input[type="text"], input[type="email"] {
      width: 100%; padding: 8px; margin-bottom: 12px;
      border: 1px solid #ccc; border-radius: 4px;
    }
    button {
      background: #4CAF50; color: white;
      padding: 10px 20px; border: none;
      border-radius: 5px; cursor: pointer;
      font-size: 15px;
    }
    button:hover { background: #45a049; }
  </style>
</head>
<body>
  <div class="form-box">
    <h2>Contact Us</h2>
    <input type="text" placeholder="Your Name">
    <input type="email" placeholder="Your Email">
    <button>Submit</button>
  </div>
</body>
</html>
```

Practical No. 17

Q. Develop a responsive portfolio webpage using Bootstrap Navbar, Grid Layout, and Progress Bars.

```
<!DOCTYPE html>
<html>
<head>
  <title>Portfolio</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
    rel="stylesheet">
</head>
<body>
  <nav class="navbar navbar-dark bg-dark">
    <div class="container">
      <span class="navbar-brand">Shreyas Portfolio</span>
    </div>
  </nav>

  <div class="container mt-4">
    <h3>About Me</h3>
    <p>Computer Engineering student at SPPU.</p>

    <h3>Skills</h3>
    <p>HTML</p>
    <div class="progress mb-2">
      <div class="progress-bar" style="width:90%">90%</div>
    </div>
    <p>Python</p>
    <div class="progress mb-2">
      <div class="progress-bar bg-success" style="width:75%">75%</div>
    </div>
    <p>Java</p>
    <div class="progress mb-2">
      <div class="progress-bar bg-warning" style="width:60%">60%</div>
    </div>
  </div>
</body>
</html>
```

Practical No. 18

Q. Write a JavaScript program to validate a signup form including password confirmation and age verification.

```
<!DOCTYPE html>
<html>
<head>
  <title>Signup Validation</title>
  <script>
    function validate(e) {
      var name = document.getElementById("name").value;
      var age = parseInt(document.getElementById("age").value);
      var pass = document.getElementById("pass").value;
      var cpass = document.getElementById("cpass").value;
      var msg = "";

      if (name == "") msg += "Name is required.\n";
      if (age < 18) msg += "Age must be 18 or above.\n";
      if (pass != cpass) msg += "Passwords do not match.\n";

      if (msg != "") { alert(msg); e.preventDefault(); }
      else alert("Signup Successful!");
    }
  </script>
</head>
<body>
  <h2>Signup Form</h2>
  <form onsubmit="validate(event)">
    Name:      <input type="text" id="name"><br><br>
    Age:       <input type="number" id="age"><br><br>
    Password:  <input type="password" id="pass"><br><br>
    Confirm Password: <input type="password" id="cpass"><br><br>
    <input type="submit" value="Sign Up">
  </form>
</body>
</html>
```

Practical No. 19

Q. Create a web page that uses JavaScript DOM methods to add and remove list items dynamically.

```
<!DOCTYPE html>
<html>
<head><title>Dynamic List</title></head>
<body>
  <h2>Dynamic List</h2>
  <input type="text" id="item" placeholder="Enter item">
  <button onclick="addItem()">Add</button>
  <button onclick="removeItem()">Remove Last</button>

  <ul id="myList">
    <li>Shreyas</li>
    <li>Yash</li>
  </ul>

  <script>
    function addItem() {
      var val = document.getElementById("item").value;
      if (val == "") { alert("Enter something!"); return; }
      var li = document.createElement("li");
      li.innerHTML = val;
      document.getElementById("myList").appendChild(li);
      document.getElementById("item").value = "";
    }

    function removeItem() {
      var list = document.getElementById("myList");
      if (list.lastChild) list.removeChild(list.lastChild);
    }
  </script>
</body>
</html>
```

Practical No. 20

Q. Write a JavaScript program to generate a multiplication table for a given number using functions.

```
<!DOCTYPE html>
<html>
<head><title>Multiplication Table</title></head>
<body>
  <h2>Multiplication Table</h2>
  Number: <input type="number" id="num" placeholder="Enter number">
  <button onclick="generate()">Generate</button>

  <div id="result"></div>

  <script>
    function generate() {
      var n    = parseInt(document.getElementById("num").value);
      var html = "<h3>Table of " + n + "</h3>";
      for (var i = 1; i <= 10; i++) {
        html += n + " x " + i + " = " + (n * i) + "<br>";
      }
      document.getElementById("result").innerHTML = html;
    }
  </script>
</body>
</html>
```

Practical No. 21

Q. Design a PHP script to display a greeting message based on the current time (morning, afternoon, evening).

```
<!DOCTYPE html>
<html>
<head><title>Greeting</title></head>
<body>
  <h2>Time-Based Greeting</h2>
  <?php
    $hour = (int) date("H");

    if ($hour >= 5 && $hour < 12) {
      $greeting = "Good Morning!";
    } elseif ($hour >= 12 && $hour < 17) {
      $greeting = "Good Afternoon!";
    } elseif ($hour >= 17 && $hour < 21) {
      $greeting = "Good Evening!";
    } else {
      $greeting = "Good Night!";
    }

    echo "<h3>" . $greeting . "</h3>";
    echo "Current Time: " . date("h:i:s A");
  ?>
</body>
</html>
```

Practical No. 22

Q. Write a PHP program to accept student registration details through a form and display them using the POST method.

```
<!DOCTYPE html>
<html>
<head><title>Student Registration</title></head>
<body>
  <h2>Student Registration</h2>
  <form method="POST">
    Name: <input type="text" name="name" required><br><br>
    Email: <input type="email" name="email" required><br><br>
    Branch: <select name="branch">
      <option>Computer Engineering</option>
      <option>Mechanical</option>
      <option>Civil</option>
    </select><br><br>
    <input type="submit" name="btn" value="Register">
  </form>
  <hr>
  <?php
    if (isset($_POST['btn'])) {
      $name = htmlspecialchars($_POST['name']);
      $email = htmlspecialchars($_POST['email']);
      $branch = htmlspecialchars($_POST['branch']);
      echo "<h3>Registration Successful!</h3>";
      echo "Name: $name <br> Email: $email <br> Branch: $branch";
    }
  ?>
</body>
</html>
```

Practical No. 23

Q. Implement a PHP program to replace words in a string, find word positions, and trim spaces.

```
<!DOCTYPE html>
<html>
<head><title>PHP String Functions</title></head>
<body>
  <h2>PHP String Functions</h2>
  <?php
    $str = " Hello World Hello PHP ";

    echo "Original: '" . $str . "'<br>";
    echo "Trimmed: '" . trim($str) . "'<br>";
    echo "Replaced: '" . str_replace("Hello","Hi",$str) . "'<br>";
    echo "Position of 'World': " . strpos(trim($str),"World") . "<br>";
    echo "Substring (6,5): '" . substr(trim($str),6,5) . "'<br>";
  ?>
</body>
</html>
```

Practical No. 24

Q. Create a PHP script to store and display customer details using multidimensional arrays.

```
<!DOCTYPE html>
<html>
<head><title>Customer Details</title></head>
<body>
  <h2>Customer Details</h2>
  <?php
    $customers = array(
      array("Shreyas", "shreyas@mail.com", "Pune"),
      array("Yash", "yash@mail.com", "Nashik"),
      array("Radha", "radha@mail.com", "Mumbai"),
      array("Krishna", "krishna@mail.com", "Nagpur"),
      array("Tejas", "tejas@mail.com", "Aurangabad")
    );

    echo "<table border='1' cellpadding='8'>";
    echo "<tr><th>Name</th><th>Email</th><th>City</th></tr>";
    foreach ($customers as $c) {
      echo "<tr><td>{$c[0]}</td><td>{$c[1]}</td><td>{$c[2]}</td></tr>";
    }
    echo "</table>";
  ?>
</body>
</html>
```

Practical No. 25

Q. Create an HTML page to display a student profile including photo, educational details, and hobbies using tables and formatting tags.

```
<!DOCTYPE html>
<html>
<head><title>Student Profile</title></head>
<body>
  <h1>Student Profile</h1>
  
  <h2>Yash Kulkarni</h2>
  <p><i>Computer Engineering Student - SPPU</i></p>

  <h3>Educational Details</h3>
  <table border="1" cellpadding="8">
    <tr><th>Degree</th><th>Institution</th><th>Year</th><th>%</th></tr>
    <tr><td>B.E.</td><td>SPPU College</td><td>2027</td><td>Pursuing</td></tr>
    <tr><td>HSC</td><td>Pune Board</td><td>2023</td><td>82%</td></tr>
    <tr><td>SSC</td><td>Pune Board</td><td>2021</td><td>90%</td></tr>
  </table>

  <h3>Hobbies</h3>
  <ul>
    <li>Cricket</li>
    <li>Coding</li>
    <li>Photography</li>
  </ul>
</body>
</html>
```

Practical No. 26

Q. Design a web page that contains an image gallery, hyperlinks to different websites, and a nested table for timetable details.

```
<!DOCTYPE html>
<html>
<head><title>Gallery and Timetable</title></head>
<body>
  <h2>Image Gallery</h2>
  
  
  

  <h2>Useful Links</h2>
  <a href="https://google.com" target="_blank">Google</a> |
  <a href="https://github.com" target="_blank">GitHub</a> |
  <a href="https://w3schools.com" target="_blank">W3Schools</a>

  <h2>Timetable (Nested Table)</h2>
  <table border="1" cellpadding="6">
    <tr><th>Day</th><th>Session 1</th><th>Session 2</th></tr>
    <tr>
      <td>Monday</td>
      <td><table border="1"><tr><td>DBMS</td></tr><tr><td>Lab</td></tr></table></td>
      <td>DM</td>
    </tr>
    <tr>
      <td>Tuesday</td>
      <td>IOT</td>
      <td><table border="1"><tr><td>COM</td></tr><tr><td>Lab</td></tr></table></td>
    </tr>
  </table>
</body>
</html>
```

Practical No. 27

Q. Create a registration form with fields for name, email, password, gender, city selection, hobbies, and submit/reset buttons.

```
<!DOCTYPE html>
<html>
<head><title>Registration Form</title></head>
<body>
  <h2>Registration Form</h2>
  <form>
    Name:    <input type="text"      name="name"   required><br><br>
    Email:   <input type="email"     name="email"  required><br><br>
    Password: <input type="password" name="pass"   required><br><br>

    Gender:<br>
    <input type="radio" name="gender" value="male">   Male
    <input type="radio" name="gender" value="female"> Female<br><br>

    City:
    <select name="city">
      <option>Pune</option>
      <option>Mumbai</option>
      <option>Nashik</option>
      <option>Nagpur</option>
    </select><br><br>

    Hobbies:<br>
    <input type="checkbox" name="h" value="cricket">   Cricket
    <input type="checkbox" name="h" value="coding">    Coding
    <input type="checkbox" name="h" value="reading">   Reading<br><br>

    <input type="submit" value="Register">
    <input type="reset"  value="Reset">
  </form>
</body>
</html>
```

Practical No. 28

Q. Apply internal, external, and inline CSS to style a webpage with navigation menu, tables, and hover effects on buttons.

```
<!DOCTYPE html>
<html>
<head>
  <title>CSS Styles</title>
  <style>
    /* Internal CSS */
    nav { background: #222; padding: 10px; }
    nav a { color: white; margin: 0 12px; text-decoration: none; }
    nav a:hover { color: orange; }
    table { border-collapse: collapse; width: 50%; margin-top: 15px; }
    th, td { border: 1px solid #999; padding: 8px; text-align: left; }
    th { background: #4CAF50; color: white; }
    .btn { padding: 8px 16px; border: none; cursor: pointer; }
    .btn:hover { background: #333; color: white; }
  </style>
</head>
<body>
  <nav>
    <a href="#">Home</a>
    <a href="#">Courses</a>
    <a href="#">Contact</a>
  </nav>

  <!-- Inline CSS -->
  <h2 style="color: darkblue; text-align: center;">Student Data</h2>

  <table>
    <tr><th>Name</th><th>Marks</th></tr>
    <tr><td>Shreyas</td><td>85</td></tr>
    <tr><td>Radha</td><td>92</td></tr>
    <tr><td>Krishna</td><td>88</td></tr>
  </table>

  <br>
  <button class="btn" style="background: lightblue;">Button 1</button>
  <button class="btn" style="background: lightgreen;">Button 2</button>
</body>
</html>
```

Practical No. 29

Q. Develop a responsive webpage using Bootstrap Grid System, Cards, Navbar, and Footer components.

```
<!DOCTYPE html>
<html>
<head>
  <title>Bootstrap Page</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
    rel="stylesheet">
</head>
<body>
  <nav class="navbar navbar-dark bg-primary">
    <div class="container">
      <span class="navbar-brand">My Website</span>
    </div>
  </nav>

  <div class="container mt-4">
    <div class="row">
      <div class="col-md-4 mb-3">
        <div class="card">
          <div class="card-body">
            <h5 class="card-title">Card 1</h5>
            <p class="card-text">This is card one content.</p>
          </div>
        </div>
      </div>
      <div class="col-md-4 mb-3">
        <div class="card">
          <div class="card-body">
            <h5 class="card-title">Card 2</h5>
            <p class="card-text">This is card two content.</p>
          </div>
        </div>
      </div>
      <div class="col-md-4 mb-3">
        <div class="card">
          <div class="card-body">
            <h5 class="card-title">Card 3</h5>
            <p class="card-text">This is card three content.</p>
          </div>
        </div>
      </div>
    </div>
  </div>

  <footer class="bg-dark text-white text-center py-3 mt-4">
    &copy; 2025 My Website. All rights reserved.
  </footer>
</body>
</html>
```

Practical No. 30

Q. Write a JavaScript program to validate a form for empty fields, password length, and proper email format.

```
<!DOCTYPE html>
<html>
<head>
  <title>Form Validation</title>
  <script>
    function validate(e) {
      var name  = document.getElementById("name").value;
      var email = document.getElementById("email").value;
      var pass  = document.getElementById("pass").value;
      var msg   = "";

      if (name == "") msg += "Name is required.\n";
      if (!/^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email))
        msg += "Invalid email format.\n";
      if (pass.length < 6) msg += "Password must be 6+ characters.\n";

      if (msg != "") { alert(msg); e.preventDefault(); }
      else alert("Form Submitted!");
    }
  </script>
</head>
<body>
  <h2>Validation Form</h2>
  <form onsubmit="validate(event)">
    Name:    <input type="text"      id="name"><br><br>
    Email:   <input type="text"      id="email"><br><br>
    Password: <input type="password" id="pass"><br><br>
    <input type="submit" value="Submit">
  </form>
</body>
</html>
```

Practical No. 31

Q. Create a web page that dynamically changes images and text using JavaScript DOM manipulation.

```
<!DOCTYPE html>
<html>
<head><title>Dynamic Image and Text</title></head>
<body>
  <h2 id="title">Original Title</h2>
  <br><br>

  <button onclick="change1()">Image 1</button>
  <button onclick="change2()">Image 2</button>
  <button onclick="reset()">Reset</button>

  <script>
    function change1() {
      document.getElementById("img").src =
        "https://via.placeholder.com/200x100?text=Image+2";
      document.getElementById("title").innerHTML = "Image Changed to 2!";
    }
    function change2() {
      document.getElementById("img").src =
        "https://via.placeholder.com/200x100?text=Image+3";
      document.getElementById("title").innerHTML = "Image Changed to 3!";
    }
    function reset() {
      document.getElementById("img").src =
        "https://via.placeholder.com/200x100?text=Image+1";
      document.getElementById("title").innerHTML = "Original Title";
    }
  </script>
</body>
</html>
```

Practical No. 32

Q. Write a JavaScript program to perform calculator operations (+, -, x, /) using functions and switch-case statements.

```
<!DOCTYPE html>
<html>
<head>
  <title>Calculator</title>
  <style>
    .box { border:2px solid #333; padding:20px; width:260px; }
    input { width:90%; padding:5px; margin-bottom:8px; }
    button{ width:45%; padding:8px; margin:3px; cursor:pointer; }
    #res { font-weight:bold; color:blue; margin-top:8px; }
  </style>
</head>
<body>
  <div class="box">
    <h3>Calculator</h3>
    <input type="number" id="a" placeholder="Number 1">
    <input type="number" id="b" placeholder="Number 2">
    <button onclick="calc('+')">+</button>
    <button onclick="calc('-')">-</button>
    <button onclick="calc('*')">x</button>
    <button onclick="calc('/')">/</button>
    <div id="res">Result: </div>
  </div>
  <script>
    function calc(op) {
      var a = parseFloat(document.getElementById("a").value);
      var b = parseFloat(document.getElementById("b").value);
      var r;
      switch(op) {
        case '+': r = a + b; break;
        case '-': r = a - b; break;
        case '*': r = a * b; break;
        case '/': r = b != 0 ? a / b : "Cannot divide by 0"; break;
      }
      document.getElementById("res").innerHTML = "Result: " + r;
    }
  </script>
</body>
</html>
```

Practical No. 33

Q. Design a PHP script to display a welcome message, current date/time, and browser information.

```
<!DOCTYPE html>
<html>
<head><title>Server Info</title></head>
<body>
  <h2>Welcome Message</h2>
  <?php
    echo "<h3 style='color:green;'>Welcome to PHP!</h3>";
    echo "Current Date: " . date("d-m-Y") . "<br>";
    echo "Current Time: " . date("h:i:s A") . "<br><br>";
    echo "<b>Browser Info:</b><br>";
    echo $_SERVER['HTTP_USER_AGENT'];
  ?>
</body>
</html>
```

Practical No. 34

Q. Write a PHP program to accept student details through a form using POST method and display the entered information in table format.

```
<!DOCTYPE html>
<html>
<head><title>Student Details</title></head>
<body>
  <h2>Student Details Form</h2>
  <form method="POST">
    Name:  <input type="text"   name="name"   required><br><br>
    Email: <input type="email"  name="email"  required><br><br>
    Branch: <input type="text"   name="branch"  required><br><br>
    Marks: <input type="number"  name="marks"   required><br><br>
    <input type="submit" name="btn" value="Submit">
  </form>
  <hr>
  <?php
    if (isset($_POST['btn'])) {
      $n = htmlspecialchars($_POST['name']);
      $e = htmlspecialchars($_POST['email']);
      $b = htmlspecialchars($_POST['branch']);
      $m = htmlspecialchars($_POST['marks']);
      echo "<h3>Student Information</h3>";
      echo "<table border='1' cellpadding='8'>";
      echo "<tr><th>Field</th><th>Value</th></tr>";
      echo "<tr><td>Name</td><td>$n</td></tr>";
      echo "<tr><td>Email</td><td>$e</td></tr>";
      echo "<tr><td>Branch</td><td>$b</td></tr>";
      echo "<tr><td>Marks</td><td>$m</td></tr>";
      echo "</table>";
    }
  ?>
</body>
</html>
```

Practical No. 35

Q. Implement a PHP program to perform string operations such as reverse, uppercase, lowercase, and substring extraction.

```
<!DOCTYPE html>
<html>
<head><title>String Operations</title></head>
<body>
  <h2>String Operations</h2>
  <?php
    $str = "Computer Engineering";
    echo "Original: " . $str . "<br>";
    echo "Uppercase: " . strtoupper($str) . "<br>";
    echo "Lowercase: " . strtolower($str) . "<br>";
    echo "Reversed: " . strrev($str) . "<br>";
    echo "Substring: " . substr($str, 0, 8) . "<br>";
    echo "Length: " . strlen($str) . "<br>";
  ?>
</body>
</html>
```

Practical No. 36

Q. Create a PHP script to store student marks in an associative array and calculate total marks and average.

```
<!DOCTYPE html>
<html>
<head><title>Marks Calculator</title></head>
<body>
  <h2>Student Marks</h2>
  <?php
    $marks = array(
      "DBMS" => 85,
      "DM"   => 78,
      "IOT"  => 82,
      "COM"  => 75,
      "ES"   => 25
    );

    $total = array_sum($marks);
    $avg   = $total / count($marks);

    echo "<table border='1' cellpadding='8'>";
    echo "<tr><th>Subject</th><th>Marks</th></tr>";
    foreach ($marks as $sub => $m) {
      echo "<tr><td>$sub</td><td>$m</td></tr>";
    }
    echo "<tr><td><b>Total</b></td><td><b>$total</b></td></tr>";
    echo "<tr><td><b>Average</b></td><td><b>$avg</b></td></tr>";
    echo "</table>";
  ?>
</body>
</html>
```

Practical No. 37

Q. Create an HTML page to display faculty profile information using headings, images, paragraphs, and lists.

```
<!DOCTYPE html>
<html>
<head><title>Faculty Profile</title></head>
<body>
  <h1>Faculty Profile</h1>
  
  <h2>Dr. Rajesh Sharma</h2>
  <p><b>Designation:</b> Assistant Professor</p>
  <p><b>Department:</b> Computer Engineering</p>
  <p><b>Email:</b> rajesh@college.edu</p>

  <h3>Subjects Taught</h3>
  <ul>
    <li>Database Management System</li>
    <li>Data Structures</li>
    <li>Web Development</li>
  </ul>

  <h3>Qualifications</h3>
  <ul>
    <li>Ph.D. - Computer Science</li>
    <li>M.E. - Computer Engineering</li>
    <li>B.E. - Computer Engineering</li>
  </ul>
</body>
</html>
```

Practical No. 38

Q. Design a web page containing product images, hyperlinks to e-commerce websites, and a product price table.

```
<!DOCTYPE html>
<html>
<head><title>Product Page</title></head>
<body>
  <h2>Our Products</h2>
  
  
  

  <h2>Shop Online</h2>
  <a href="https://www.amazon.in" target="_blank">Amazon</a> |
  <a href="https://www.flipkart.com" target="_blank">Flipkart</a> |
  <a href="https://www.myntra.com" target="_blank">Myntra</a>

  <h2>Price List</h2>
  <table border="1" cellpadding="8">
    <tr><th>Product</th><th>Brand</th><th>Price (Rs.)</th></tr>
    <tr><td>Laptop</td><td>Dell</td><td>55,000</td></tr>
    <tr><td>Phone</td><td>Samsung</td><td>18,000</td></tr>
    <tr><td>Tablet</td><td>Apple</td><td>45,000</td></tr>
  </table>
</body>
</html>
```

Practical No. 39

Q. Create an online student admission form with fields for student name, email address, gender, branch selection, hobbies, and submit button.

```
<!DOCTYPE html>
<html>
<head><title>Admission Form</title></head>
<body>
  <h2>Student Admission Form</h2>
  <form>
    Student Name: <input type="text" name="name" required><br><br>
    Email:        <input type="email" name="email" required><br><br>

    Gender:<br>
    <input type="radio" name="gender" value="male">   Male
    <input type="radio" name="gender" value="female"> Female<br><br>

    Branch:
    <select name="branch">
      <option>Computer Engineering</option>
      <option>Mechanical Engineering</option>
      <option>Civil Engineering</option>
      <option>Electronics</option>
    </select><br><br>

    Hobbies:<br>
    <input type="checkbox" value="cricket">   Cricket
    <input type="checkbox" value="coding">    Coding
    <input type="checkbox" value="music">     Music<br><br>

    <input type="submit" value="Submit Application">
  </form>
</body>
</html>
```

Practical No. 40

Q. Apply CSS styles using inline, internal, and external CSS to create a styled login form and footer section.

```
<!DOCTYPE html>
<html>
<head>
  <title>Login Form</title>
  <style>
    /* Internal CSS */
    body { font-family: Arial; background: #f5f5f5; }
    .login-box {
      background: white; width: 320px; margin: 50px auto;
      padding: 30px; border-radius: 8px;
      box-shadow: 0 0 10px rgba(0,0,0,0.15);
    }
    input[type="text"], input[type="password"] {
      width: 100%; padding: 8px; margin-bottom: 12px;
      border: 1px solid #ccc; border-radius: 4px;
    }
    .btn-login {
      width: 100%; padding: 10px; background: #007bff;
      color: white; border: none; border-radius: 4px;
      cursor: pointer;
    }
    footer {
      text-align: center; padding: 15px;
      background: #333; color: white; margin-top: 20px;
    }
  </style>
</head>
<body>
  <div class="login-box">
    <!-- Inline CSS heading -->
    <h2 style="text-align:center; color:#007bff;">Login</h2>
    <input type="text"      placeholder="Username">
    <input type="password" placeholder="Password">
    <button class="btn-login">Login</button>
  </div>
  <footer>
    &copy; 2025 College Portal. All Rights Reserved.
  </footer>
</body>
</html>
```

Practical No. 41

Q. Develop a responsive hospital website homepage using Bootstrap Carousel, Cards, and Grid System.

```
<!DOCTYPE html>
<html>
<head>
  <title>Hospital Website</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
    rel="stylesheet">
</head>
<body>
  <nav class="navbar navbar-dark bg-danger">
    <div class="container">
      <span class="navbar-brand fw-bold">City Hospital</span>
    </div>
  </nav>

  <!-- Carousel -->
  <div id="carouselEx" class="carousel slide" data-bs-ride="carousel">
    <div class="carousel-inner">
      <div class="carousel-item active">
        <div class="bg-info text-white text-center p-5">
          <h2>Quality Healthcare</h2></div>
        </div>
      <div class="carousel-item">
        <div class="bg-success text-white text-center p-5">
          <h2>24/7 Emergency Services</h2></div>
        </div>
      </div>
    <div>
      <button class="carousel-control-prev" data-bs-target="#carouselEx"
        data-bs-slide="prev">
        <span class="carousel-control-prev-icon"></span>
      </button>
      <button class="carousel-control-next" data-bs-target="#carouselEx"
        data-bs-slide="next">
        <span class="carousel-control-next-icon"></span>
      </button>
    </div>
  </div>

  <div class="container mt-4">
    <div class="row">
      <div class="col-md-4 mb-3">
        <div class="card border-danger">
          <div class="card-body">
            <h5 class="card-title">Cardiology</h5>
            <p>Expert heart care department.</p>
          </div>
        </div>
      </div>
      <div class="col-md-4 mb-3">
        <div class="card border-primary">
          <div class="card-body">
            <h5 class="card-title">Orthopaedics</h5>
            <p>Advanced bone and joint care.</p>
          </div>
        </div>
      </div>
      <div class="col-md-4 mb-3">
        <div class="card border-success">
          <div class="card-body">
            <h5 class="card-title">Neurology</h5>
            <p>Specialist brain and nerve care.</p>
          </div>
        </div>
      </div>
    </div>
  </div>
```

```
    </div>
  </div>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js">
</script>
</body>
</html>
```

Practical No. 42

Q. Write a JavaScript program to validate a student login form including email format, password confirmation, and empty field checking.

```
<!DOCTYPE html>
<html>
<head>
  <title>Login Validation</title>
  <script>
    function validate(e) {
      var email = document.getElementById("email").value;
      var pass = document.getElementById("pass").value;
      var msg = "";

      if (email == "") msg += "Email is required.\n";
      else if (!/^[\s@]+@[\s@]+\.[\s@]+$/ .test(email))
        msg += "Invalid email format.\n";
      if (pass == "") msg += "Password is required.\n";

      if (msg != "") { alert(msg); e.preventDefault(); }
      else alert("Login Successful!");
    }
  </script>
</head>
<body>
  <h2>Student Login</h2>
  <form onsubmit="validate(event)">
    Email:   <input type="text"      id="email"><br><br>
    Password: <input type="password" id="pass"><br><br>
    <input type="submit" value="Login">
  </form>
</body>
</html>
```

Practical No. 43

Q. Create a web page that changes font size, text color, and image dynamically using JavaScript DOM manipulation.

```
<!DOCTYPE html>
<html>
<head><title>Dynamic Style Changes</title></head>
<body>
  <h2 id="txt">This is a Sample Text</h2>
  <br><br>

  <button onclick="bigFont()">Big Font</button>
  <button onclick="colorRed()">Red Text</button>
  <button onclick="changeImg()">Change Image</button>
  <button onclick="resetAll()">Reset</button>

  <script>
    function bigFont() {
      document.getElementById("txt").style.fontSize = "32px";
    }
    function colorRed() {
      document.getElementById("txt").style.color = "red";
    }
    function changeImg() {
      document.getElementById("pic").src =
        "https://via.placeholder.com/150/FF0000?text=New";
    }
    function resetAll() {
      var t = document.getElementById("txt");
      t.style.fontSize = "";
      t.style.color = "";
      document.getElementById("pic").src =
        "https://via.placeholder.com/150";
    }
  </script>
</body>
</html>
```

Practical No. 44

Q. Write a JavaScript program to calculate simple interest, compound interest, and percentage using buttons and functions.

```
<!DOCTYPE html>
<html>
<head>
  <title>Interest Calculator</title>
  <style>
    input { margin-bottom: 8px; padding: 5px; width: 200px; }
    button { padding: 8px 15px; margin: 4px; cursor: pointer; }
    #res { font-weight: bold; color: darkblue; margin-top: 10px; }
  </style>
</head>
<body>
  <h2>Interest Calculator</h2>
  Principal (P): <input type="number" id="p"><br>
  Rate (R %):   <input type="number" id="r"><br>
  Time (T yrs): <input type="number" id="t"><br>

  <button onclick="calcSI()">Simple Interest</button>
  <button onclick="calcCI()">Compound Interest</button>
  <button onclick="calcPct()">Percentage</button>

  <div id="res"></div>

  <script>
    function calcSI() {
      var p=parseFloat(document.getElementById("p").value);
      var r=parseFloat(document.getElementById("r").value);
      var t=parseFloat(document.getElementById("t").value);
      var si = (p * r * t) / 100;
      document.getElementById("res").innerHTML =
        "Simple Interest = Rs. " + si.toFixed(2);
    }
    function calcCI() {
      var p=parseFloat(document.getElementById("p").value);
      var r=parseFloat(document.getElementById("r").value);
      var t=parseFloat(document.getElementById("t").value);
      var ci = p * Math.pow((1 + r/100), t) - p;
      document.getElementById("res").innerHTML =
        "Compound Interest = Rs. " + ci.toFixed(2);
    }
    function calcPct() {
      var p=parseFloat(document.getElementById("p").value);
      var r=parseFloat(document.getElementById("r").value);
      document.getElementById("res").innerHTML =
        r + "% of " + p + " = " + ((p * r) / 100).toFixed(2);
    }
  </script>
</body>
</html>
```

Practical No. 45

Q. Design a PHP script to display student information and browser details using predefined server variables.

```
<!DOCTYPE html>
<html>
<head><title>Student Info</title></head>
<body>
  <h2>Student Information</h2>
  <?php
    $name    = "Shreyas Patil";
    $roll    = "22CE101";
    $branch  = "Computer Engineering";
    $sem     = "Semester IV";

    echo "Name:  $name  <br>";
    echo "Roll No: $roll <br>";
    echo "Branch: $branch <br>";
    echo "Semester: $sem <br><br>";

    echo "<h3>Browser & Server Details</h3>";
    echo "IP Address: " . $_SERVER['REMOTE_ADDR'] . "<br>";
    echo "Browser: " . $_SERVER['HTTP_USER_AGENT'] . "<br>";
    echo "Server: " . $_SERVER['SERVER_SOFTWARE'] . "<br>";
    echo "Current Date: " . date("d-m-Y") . "<br>";
  ?>
</body>
</html>
```

Practical No. 46

Q. Write a PHP program to accept user feedback using POST method and display the entered details in formatted output.

```
<!DOCTYPE html>
<html>
<head><title>Feedback Form</title></head>
<body>
  <h2>User Feedback</h2>
  <form method="POST">
    Name:   <input type="text" name="name"   required><br><br>
    Email:  <input type="email" name="email"  required><br><br>
    Rating:
    <select name="rating">
      <option>Excellent</option>
      <option>Good</option>
      <option>Average</option>
      <option>Poor</option>
    </select><br><br>
    Comments:<br>
    <textarea name="comments" rows="4" cols="40"></textarea><br><br>
    <input type="submit" name="btn" value="Submit Feedback">
  </form>
  <hr>
  <?php
    if (isset($_POST['btn'])) {
      $n = htmlspecialchars($_POST['name']);
      $e = htmlspecialchars($_POST['email']);
      $r = htmlspecialchars($_POST['rating']);
      $c = htmlspecialchars($_POST['comments']);
      echo "<h3>Feedback Received!</h3>";
      echo "<b>Name:</b> $n <br>";
      echo "<b>Email:</b> $e <br>";
      echo "<b>Rating:</b> $r <br>";
      echo "<b>Comments:</b> $c <br>";
    }
  ?>
</body>
</html>
```

Practical No. 47

Q. Implement a PHP program for array operations such as sorting, searching, and finding maximum value.

```
<!DOCTYPE html>
<html>
<head><title>Array Operations</title></head>
<body>
  <h2>Array Operations</h2>
  <?php
    $marks = array(75, 90, 55, 88, 62, 45, 95);

    echo "Original Array: " . implode(" ", $marks) . "<br>";

    sort($marks);
    echo "Sorted (Asc): " . implode(" ", $marks) . "<br>";

    rsort($marks);
    echo "Sorted (Desc): " . implode(" ", $marks) . "<br>";

    echo "Max Value: " . max($marks) . "<br>";
    echo "Min Value: " . min($marks) . "<br>";

    $search = 88;
    $found = in_array($search, $marks);
    echo "Search $search: " . ($found ? "Found" : "Not Found") . "<br>";
    echo "Total Elements: " . count($marks) . "<br>";
  ?>
</body>
</html>
```

Practical No. 48

Q. Create a PHP script to store product names and prices in an associative array and display them in tabular format.

```
<!DOCTYPE html>
<html>
<head><title>Product List</title></head>
<body>
  <h2>Product Price List</h2>
  <?php
    $products = array(
      "Laptop"      => 55000,
      "Smartphone" => 18000,
      "Tablet"      => 25000,
      "Headphones" => 3500,
      "Keyboard"    => 1200
    );

    echo "<table border='1' cellpadding='8'>";
    echo "<tr><th>Product</th><th>Price (Rs.)</th></tr>";
    foreach ($products as $name => $price) {
      echo "<tr><td>$name</td><td>$price</td></tr>";
    }
    echo "</table>";
  ?>
</body>
</html>
```

Practical No. 49

Q. Create an HTML page to display a tourist place description using headings, paragraphs, images, and unordered lists.

```
<!DOCTYPE html>
<html>
<head><title>Tourist Place</title></head>
<body>
  <h1>Mahabaleshwar - Queen of Hill Stations</h1>
  

  <h2>About</h2>
  <p>Mahabaleshwar is a hill station in Satara district, Maharashtra.
    It is famous for strawberries, scenic viewpoints, and pleasant weather.</p>

  <h2>Top Attractions</h2>
  <ul>
    <li>Venna Lake</li>
    <li>Elephant Head Point</li>
    <li>Pratapgad Fort</li>
    <li>Mapro Garden</li>
    <li>Arthur's Seat</li>
  </ul>

  <h2>How to Reach</h2>
  <p>Nearest city: <b>Pune</b> (120 km). Best reached by road.</p>
</body>
</html>
```

Practical No. 50

Q. Design a web page containing a photo gallery, hyperlinks to educational websites, and a weekly class schedule using tables.

```
<!DOCTYPE html>
<html>
<head><title>Class Schedule</title></head>
<body>
  <h2>Photo Gallery</h2>
  
  
  

  <h2>Educational Websites</h2>
  <a href="https://nptel.ac.in" target="_blank">NPTEL</a> |
  <a href="https://coursera.org" target="_blank">Coursera</a> |
  <a href="https://geeksforgeeks.org" target="_blank">GeeksforGeeks</a>

  <h2>Weekly Schedule</h2>
  <table border="1" cellpadding="8">
    <tr><th>Day</th><th>9-10</th><th>10-11</th><th>11-12</th><th>2-4 (Lab)</th></tr>
    <tr><td>Monday</td><td>DBMS</td><td>DM</td><td>IOT</td><td>DBMS Lab</td></tr>
    <tr><td>Tuesday</td><td>COM</td><td>DBMS</td><td>DM</td><td>WD Lab</td></tr>
    <tr><td>Wednesday</td><td>IOT</td><td>COM</td><td>DBMS</td><td>-</td></tr>
    <tr><td>Thursday</td><td>DM</td><td>IOT</td><td>COM</td><td>COM Lab</td></tr>
    <tr><td>Friday</td><td>DBMS</td><td>DM</td><td>IOT</td><td>-</td></tr>
  </table>
</body>
</html>
```

Practical No. 51

Q. Create an online job application form with fields for applicant name, email, mobile number, qualification, experience, and submit button.

```
<!DOCTYPE html>
<html>
<head><title>Job Application</title></head>
<body>
  <h2>Online Job Application</h2>
  <form>
    Applicant Name:  <input type="text"   name="name"   required><br><br>
    Email:           <input type="email"  name="email"  required><br><br>
    Mobile Number:   <input type="tel"    name="mobile" required><br><br>

    Qualification:
    <select name="qual">
      <option>B.E.</option>
      <option>M.E.</option>
      <option>MBA</option>
      <option>Diploma</option>
    </select><br><br>

    Experience (years): <input type="number" name="exp" min="0"><br><br>

    <input type="submit" value="Apply Now">
    <input type="reset"  value="Clear">
  </form>
</body>
</html>
```

Practical No. 52

Q. Apply CSS styles using inline, internal, and external methods to design a registration form and styled alert messages.

```
<!DOCTYPE html>
<html>
<head>
  <title>Styled Registration</title>
  <style>
    /* Internal CSS */
    body { font-family: Arial; background: #eef; }
    .form-container {
      background: white; width: 350px;
      padding: 25px; margin: 30px auto;
      border-radius: 8px;
      box-shadow: 0 0 10px rgba(0,0,0,0.1);
    }
    input[type="text"], input[type="email"] {
      width: 100%; padding: 8px; margin-bottom: 10px;
      border: 1px solid #ccc; border-radius: 4px;
    }
    .btn { background: #5cb85c; color: white; padding: 10px 20px;
      border: none; border-radius: 4px; cursor: pointer; }
    .btn:hover { background: #449d44; }
    .alert-box {
      background: #d4edda; border: 1px solid #c3e6cb;
      padding: 10px; border-radius: 4px; margin-top: 10px;
    }
  </style>
</head>
<body>
  <div class="form-container">
    <!-- Inline CSS heading -->
    <h2 style="color:#5cb85c; text-align:center;">Register</h2>
    <input type="text" placeholder="Full Name">
    <input type="email" placeholder="Email Address">
    <button class="btn">Register</button>
    <div class="alert-box">
      Registration form ready. Fill details and submit.
    </div>
  </div>
</body>
</html>
```

Practical No. 53

Q. Develop a responsive restaurant website homepage using Bootstrap Navbar, Carousel, Cards, and Grid Layout.

```
<!DOCTYPE html>
<html>
<head>
  <title>Restaurant Website</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
    rel="stylesheet">
</head>
<body>
  <nav class="navbar navbar-dark bg-dark">
    <div class="container">
      <span class="navbar-brand fw-bold">Spice Garden</span>
    </div>
  </nav>

  <div id="menuCarousel" class="carousel slide" data-bs-ride="carousel">
    <div class="carousel-inner">
      <div class="carousel-item active">
        <div class="bg-warning text-dark text-center p-5">
          <h2>Delicious Food Awaits!</h2></div>
        </div>
      <div class="carousel-item">
        <div class="bg-success text-white text-center p-5">
          <h2>Fresh Ingredients Daily</h2></div>
        </div>
      </div>
    <div class="carousel-control-prev" data-bs-target="#menuCarousel"
      data-bs-slide="prev">
      <span class="carousel-control-prev-icon"></span>
    </div>
    <div class="carousel-control-next" data-bs-target="#menuCarousel"
      data-bs-slide="next">
      <span class="carousel-control-next-icon"></span>
    </div>
  </div>

  <div class="container mt-4">
    <h3 class="text-center">Our Menu</h3>
    <div class="row">
      <div class="col-md-4 mb-3">
        <div class="card shadow">
          <div class="card-body text-center">
            <h5>Starters</h5>
            <p>Soup, Salad, Spring Rolls</p>
          </div>
        </div>
      </div>
      <div class="col-md-4 mb-3">
        <div class="card shadow">
          <div class="card-body text-center">
            <h5>Main Course</h5>
            <p>Biriyani, Dal Tadka, Paneer</p>
          </div>
        </div>
      </div>
      <div class="col-md-4 mb-3">
        <div class="card shadow">
          <div class="card-body text-center">
            <h5>Desserts</h5>
            <p>Gulab Jamun, Ice Cream</p>
          </div>
        </div>
      </div>
    </div>
  </div>
```

```
        </div>
    </div>
</div>
</div>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js">
</script>
</body>
</html>
```

Practical No. 54

Q. Write a JavaScript program to validate a feedback form including email validation, comment length, and mandatory field checking.

```
<!DOCTYPE html>
<html>
<head>
  <title>Feedback Validation</title>
  <script>
    function validate(e) {
      var name      = document.getElementById("name").value;
      var email     = document.getElementById("email").value;
      var comments  = document.getElementById("comments").value;
      var msg       = "";

      if (name == "") msg += "Name is required.\n";
      if (!/^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email))
        msg += "Invalid email format.\n";
      if (comments.length < 10)
        msg += "Comments must be at least 10 characters.\n";

      if (msg != "") { alert(msg); e.preventDefault(); }
      else alert("Feedback submitted successfully!");
    }
  </script>
</head>
<body>
  <h2>Feedback Form</h2>
  <form onsubmit="validate(event)">
    Name:      <input type="text" id="name"><br><br>
    Email:     <input type="text" id="email"><br><br>
    Comments:<br>
    <textarea id="comments" rows="4" cols="35"></textarea><br><br>
    <input type="submit" value="Submit Feedback">
  </form>
</body>
</html>
```

Practical No. 55

Q. Create a web page that changes text style, image size, and background color dynamically using JavaScript events.

```
<!DOCTYPE html>
<html>
<head><title>Dynamic Events</title></head>
<body>
  <h2 id="heading">Sample Heading</h2>
  <br><br>

  <button onclick="boldText()">Bold Text</button>
  <button onclick="bigImage()">Big Image</button>
  <button onclick="changeBg()">Yellow Background</button>
  <button onclick="resetAll()">Reset</button>

  <script>
    function boldText() {
      document.getElementById("heading").style.fontWeight = "bold";
      document.getElementById("heading").style.color      = "darkblue";
    }
    function bigImage() {
      document.getElementById("img").width = 250;
    }
    function changeBg() {
      document.body.style.backgroundColor = "yellow";
    }
    function resetAll() {
      var h = document.getElementById("heading");
      h.style.fontWeight = "normal";
      h.style.color      = "black";
      document.getElementById("img").width = 150;
      document.body.style.backgroundColor = "white";
    }
  </script>
</body>
</html>
```

Practical No. 56

Q. Write a JavaScript program to perform area calculations of circle, rectangle, and triangle using functions and buttons.

```
<!DOCTYPE html>
<html>
<head>
  <title>Area Calculator</title>
  <style>
    input { margin-bottom: 8px; padding: 5px; width: 180px; }
    button { padding: 8px 14px; margin: 4px; cursor: pointer; }
    #res { font-weight: bold; color: darkgreen; margin-top: 10px; }
  </style>
</head>
<body>
  <h2>Area Calculator</h2>
  Value 1: <input type="number" id="v1" placeholder="Radius / Length / Base"><br>
  Value 2: <input type="number" id="v2" placeholder="Width / Height (if needed)"><br>

  <button onclick="circleArea()">Circle Area</button>
  <button onclick="rectArea()">Rectangle Area</button>
  <button onclick="triArea()">Triangle Area</button>

  <div id="res"></div>

  <script>
    function circleArea() {
      var r = parseFloat(document.getElementById("v1").value);
      document.getElementById("res").innerHTML =
        "Circle Area = " + (Math.PI * r * r).toFixed(2);
    }
    function rectArea() {
      var l = parseFloat(document.getElementById("v1").value);
      var w = parseFloat(document.getElementById("v2").value);
      document.getElementById("res").innerHTML =
        "Rectangle Area = " + (l * w).toFixed(2);
    }
    function triArea() {
      var b = parseFloat(document.getElementById("v1").value);
      var h = parseFloat(document.getElementById("v2").value);
      document.getElementById("res").innerHTML =
        "Triangle Area = " + (0.5 * b * h).toFixed(2);
    }
  </script>
</body>
</html>
```

Practical No. 57

Q. Design a PHP script to display customer details and current date/time information from the server.

```
<!DOCTYPE html>
<html>
<head><title>Customer Details</title></head>
<body>
  <h2>Customer Details</h2>
  <?php
    $customers = array(
      array("name" => "Shreyas", "email" => "shreyas@mail.com", "city" => "Pune"),
      array("name" => "Radha", "email" => "radha@mail.com", "city" => "Mumbai"),
      array("name" => "Krishna", "email" => "krishna@mail.com", "city" => "Nashik")
    );

    echo "<table border='1' cellpadding='8'>";
    echo "<tr><th>Name</th><th>Email</th><th>City</th></tr>";
    foreach ($customers as $c) {
      echo "<tr><td>{$c['name']}</td><td>{$c['email']}</td><td>{$c['city']}</td></tr>";
    }
    echo "</table><br>";

    echo "<b>Server Date:</b> " . date("d-m-Y") . "<br>";
    echo "<b>Server Time:</b> " . date("h:i:s A") . "<br>";
  ?>
</body>
</html>
```

Practical No. 58

Q. Write a PHP program to accept student registration details using POST method and display registration success message.

```
<!DOCTYPE html>
<html>
<head><title>Student Registration</title></head>
<body>
  <h2>Student Registration Form</h2>
  <form method="POST">
    Name:    <input type="text"      name="name"   required><br><br>
    Email:   <input type="email"     name="email"  required><br><br>
    Password: <input type="password" name="pass"   required><br><br>
    Branch:
    <select name="branch">
      <option>Computer Engineering</option>
      <option>Mechanical</option>
      <option>Civil</option>
    </select><br><br>
    <input type="submit" name="btn" value="Register">
  </form>
  <hr>
  <?php
    if (isset($_POST['btn'])) {
      $name  = htmlspecialchars($_POST['name']);
      $branch = htmlspecialchars($_POST['branch']);
      echo "<h3 style='color:green;'>Registration Successful!</h3>";
      echo "Welcome <b>$name</b>! You are registered in <b>$branch</b>.";
    }
  ?>
</body>
</html>
```

Practical No. 59

Q. Implement a PHP program for numeric operations such as finding factorial, palindrome number, and reverse number.

```
<!DOCTYPE html>
<html>
<head><title>Numeric Operations</title></head>
<body>
  <h2>Numeric Operations</h2>
  <?php
    $num = 5;

    // Factorial
    $fact = 1;
    for ($i = 1; $i <= $num; $i++) $fact *= $i;
    echo "Factorial of $num = $fact <br>";

    // Reverse
    $rev = (int) strrev((string) $num);
    echo "Reverse of $num = $rev <br>";

    // Palindrome check
    if ($num == $rev)
      echo "$num is a Palindrome <br>";
    else
      echo "$num is NOT a Palindrome <br>";

    // Check with another number
    $num2 = 121;
    $rev2 = (int) strrev((string) $num2);
    echo "<br>Reverse of $num2 = $rev2 <br>";
    if ($num2 == $rev2)
      echo "$num2 is a Palindrome <br>";
    else
      echo "$num2 is NOT a Palindrome <br>";
  ?>
</body>
</html>
```

Practical No. 60

Q. Create a PHP script to store book names and authors in an associative array and display them in table format.

```
<!DOCTYPE html>
<html>
<head><title>Book List</title></head>
<body>
  <h2>Book List</h2>
  <?php
    $books = array(
      "Let Us C"          => "Yashavant Kanetkar",
      "Head First Java"   => "Kathy Sierra",
      "DBMS Concepts"    => "Silberschatz",
      "The Lean Startup"  => "Eric Ries",
      "Clean Code"       => "Robert C. Martin"
    );

    echo "<table border='1' cellpadding='8'>";
    echo "<tr><th>Book Name</th><th>Author</th></tr>";
    foreach ($books as $book => $author) {
      echo "<tr><td>$book</td><td>$author</td></tr>";
    }
    echo "</table>";
  ?>
</body>
</html>
```